Setup and Configure a LAN (Local Area network) using a switch and ethernet Cables.

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Aim :

Setup and configure a LAN (Local Area network) using a switch and ethernet cables in your lab.

What is a LAN ?

A Local Area Network (LAN) refers to a network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers and internet access. LAN connects devices to promote collaboration and transfer information between users, such as computers, printers, servers and switches.

How to set up a LAN

Step 1 :

plan and design an appropriate network topology taking into account network requirements and equipment location.

Step 2 :

You can take 4 computers, a switch with 8, 16, (or) 24 ports which is sufficient for networks of these sizes and 4 ethernet cables.

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Step 3:

Connect your computers to network switch via an ethernet cable, which is as simple as plugging one end of the ethernet cable into your computer and the other end into your network switch.

Step 4:

Assign Ip address to your pc's

1. Log on to the client computer as Administrator or as Owner.
2. click Network and Internet Connections.
3. Right click Local Area connection / Ethernet → Go to properties → Select Internet protocol (TCP / IP v4) → ~~client~~ click on properties → select use the following ip address option and assign ipaddress.

Step 5:

Configure a network switch

1. Connect your computer to the switch.
2. Log in to the web interface.
3. Configure basic settings
4. Assign Ip address as; 10.1.1.5, subnet mask 255.0.0.0.

Step 6:

Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

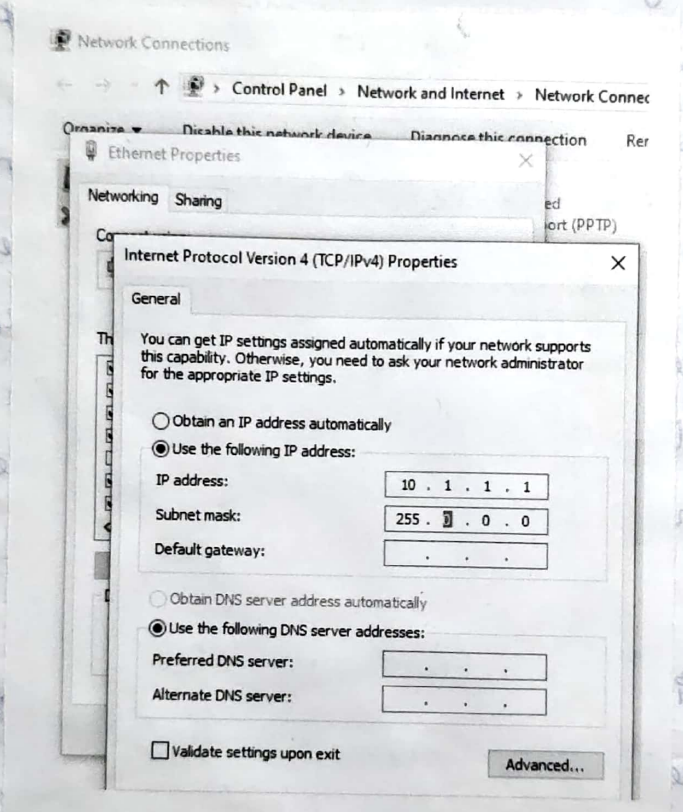
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Step 7 :-

Select a folder → go to properties → click sharing tab → share it with everyone on the same LAN.

Step 8 :-

Try to access the shared folder from others computers of the network.



Result :

Thus, the setup and configure a LAN using switch & Ethernet cable was successfully Completed.

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